KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT REPORT

LINE FOLLOWING ROBOT

A first-year project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering

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CERTIFICATION

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(Signature)	(Name)	(Date)

ABSTRACT

An autonomous robot that follows a specific trajectory or path which could be magnetic, white, or black line is known as a line following robot. An IR proxy sensor, a DC motor, a motor driver, a voltage regulator, sensor modules, jumper wires, wheels, etc. are some of the components of this robot. This is a straightforward electronics project in which a robot uses an IR proxy sensor to feel the path in its direction and then follows it. The circuit that drives our IR sensors and motor driver is equipped with a rechargeable battery and a voltage regulator.

Typically, this robot is made to increase worker comfort while lowering the risk of human labor. Due to its increased efficiency, cheaper labor costs, and capacity to function in harsh conditions, it also offers a competitive advantage over competing technologies. This robot can be used in a number of industries, including those related to home appliances, healthcare, hospitality, and so on.

ACKNOWLEDGEMENT

We would like to extend a very warm thanks to our project coordinator, Mr.Pramish Shrestha, for approving our project, "Line Following Robot," as our academic project for the duration of this semester. We would also like to express our profound gratitude to our supervisor , for his governance and guidance, because of which the completion of our project was possible. At last, we would also like to share our sincere gratitude to the entire faculty of the Department of Electrical and Electronics Engineering for the help and support during this project.

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ABBREVIATIONS

Symbols	Description
LFR	Line Following Robot
IR	Infrared Ray
DC	Direct Current
PCB	Printed Circuit Board
LCD	Liquid Crystal Display
AC	Alternating Current
LED	Light Emitting Diode
RX	Receiver

TX	Transmitter
OP	Operational

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CHAPTER I: INTRODUCTION

1.1 Background

various applications and purposes in different fields, as it can be utilized for a wide range of home and commercial tasks, including transportation, floor cleaning, delivery, and the carrying of items.

1.2 Problem Definition

This project may be used in various ways like carrying goods, floor cleaning, delivery services, transportation, etc. The line follower robot can also be used as a transportation wheel for manufacturing. It can also be used in cars, taxis, and buses to create an autonomous transport system. This solves the problem for physically challenged people. In restaurants, it can also help in many sections such as food servers to the customers, order taking jobs, etc which can be easily done. These can be used in public places like shopping malls, museums, etc to provide path guidance.

1.3 Objective

Training ourselves to design, develop, and implement in the field of electronics is one of the main goals of this project.

Meanwhile, some significant abilities listed by students are:

1. To create a robot that follows lines.
2. To learn about electronic components and develop an attitude towards learning and a desire to collaborate with others.

3. To develop logical thinking, which fosters creativity and invention and improves problem-solving abilities.

1.4 Methodology

1. Literature Survey:

The idea of the project was well defined after the detail literature survey through books, periodical, journal, magazine websites.

2. Circuit Design, Test and Simulation:

The circuit was designed and simulation on the breadboard, and testing was done correspondingly.

3. Circuit Assembly in PCB:

PCB was designed in proteus and was successfully simulated.

4. Assembly of Final Product:

After the finishing of electrical and electronics component, the stable, reliable and flexible mechanical design and fabrication was completed. Finally system was tested and problems encountered was omitted

1.5 Limitation

A few restrictions apply to this project. Here is a list of them:

1. Without a microprocessor, the robot won't be able to move backwards.

2. The robot can't make sudden turns.

1.6 Organization Of Report

There are five chapters in this report:

1. A broad summary of the project, including its goals, the problem statement, and the limitations section, is given in Chapter 1.

2. Chapter 2 surveys the literature and describes the issues that the earlier researchers ran across and the solutions they came up with.

3. The arrangement of components required for executing the project is described in depth in Chapter 3, together with the block and schematic diagrams and a thorough explanation of the component list.

4. The results portion includes Chapter 4, which details the outcome attained upon completion of the undertaking.

5. A quick summary of the project's important findings is provided in Chapter 5.

1.7 Summary

Line Follower Robot is a simple autonomously guided robot that follows a line drawn on the ground to either detect a dark line on a white surface or a white line on a dark. It is designed to move and follow the plotted line. It makes turns of varying degrees and is insensitive to environmental factors such as lighting and noise. This project enhances the

interdisciplinary approach to mechanical, electrical and electronics skills. Robots are created to lessen the stress on humans. This project is somehow designed to perform similar task.

CHAPTER 2: LITERATURE SURVEY

In one study focused on library inventory management, a Line Following Robot was designed for organizing bookshelves [1]. Equipped with sensor-operated motors, the robot navigates a predetermined line path corresponding to the arrangement of library bookshelves. It incorporates a barcode reader to collect data from vertically arranged books, comparing the decoded barcode information with user-inputted search data. When reaching the targeted book, the robot communicates its location to the librarian or visitor, streamlining book retrieval processes and minimizing manual labor for library staff.

Another project, labeled "Smart and Intelligent Line Follower Robot with Obstacle Detection," employed components such as IR sensors, Arduino, ultrasonic sensors, and motor drivers [2]. This robot possesses the ability to detect visual lines, regardless of color or contrast, and is enhanced with obstacle detection capabilities. Unlike conventional line followers, it avoids damage by halting its motion when encountering obstacles until the obstruction is removed or the path is altered, providing added durability.

A Line Follower Robot utilized components such as Arduino Uno R3, Digital IR sensor array, Adafruit Motor Shield, DC motors, Chassis board, and power supply [3]. Tailored for industrial purposes, this autonomous robot relies on an array of five IR sensors to follow a line, with readings transmitted to an Arduino for control. Its application is particularly beneficial for transporting heavy and hazardous products in industrial settings.

In another project, a Line Follower Robot featured IR sensors, a microcontroller, motor driver, camera, alarm circuit, and LCD display [4]. This project integrated patient-activated switches to guide the robot to patients, triggering an alarm in emergency situations. Additionally, the inclusion of a GSM module allows users to call for medical assistance from doctors and nurses.

In a more recreational context, a Line Follower Robot was developed for entertainment purposes [5]. This robot operates in two modes: Line Following Module and Service Mode. In Line-Following Mode, it autonomously follows a black line on a suitable surface. In Service Mode, it can be manually controlled in various directions using a remote controller. Two permanent magnet DC motors, H-Bridge circuits, and rechargeable batteries, either through AC supply or a solar panel, facilitate the robot's movements and power source.

CHAPTER 3: PROJECT METHODOLOGY

3.1 Block Diagram

The Line Follow robot is a straightforward electronics project in which a robot uses an IR sensor to sense the path put out in front of it and then uses motor wheels to follow that path. By adjusting the amount of current that passes through the L293D Motor, the robot is even capable of turning. The project is entirely constructed from electronic components; micro-controllers of any kind are not utilized.

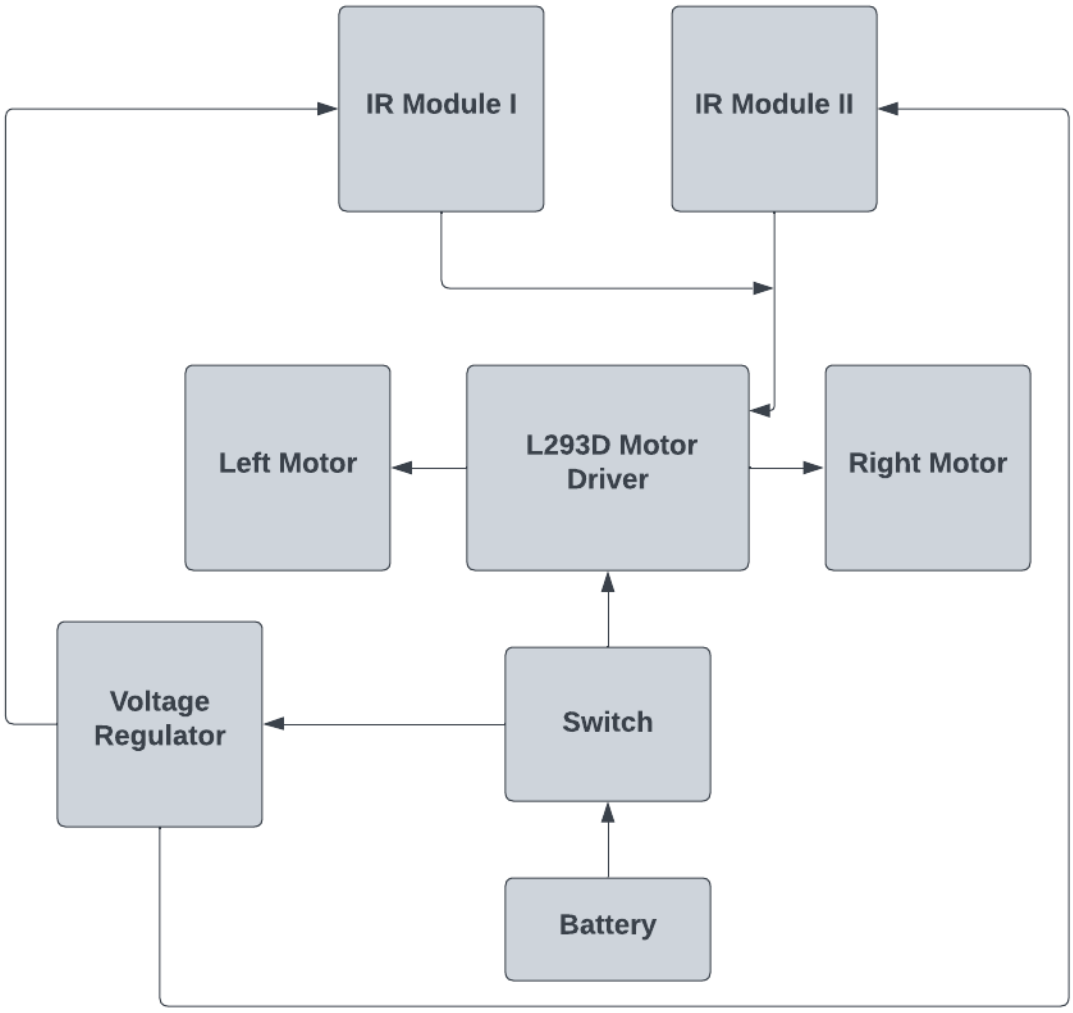


Figure 3.1 Block diagram of the project

3.2 Schematic Diagram

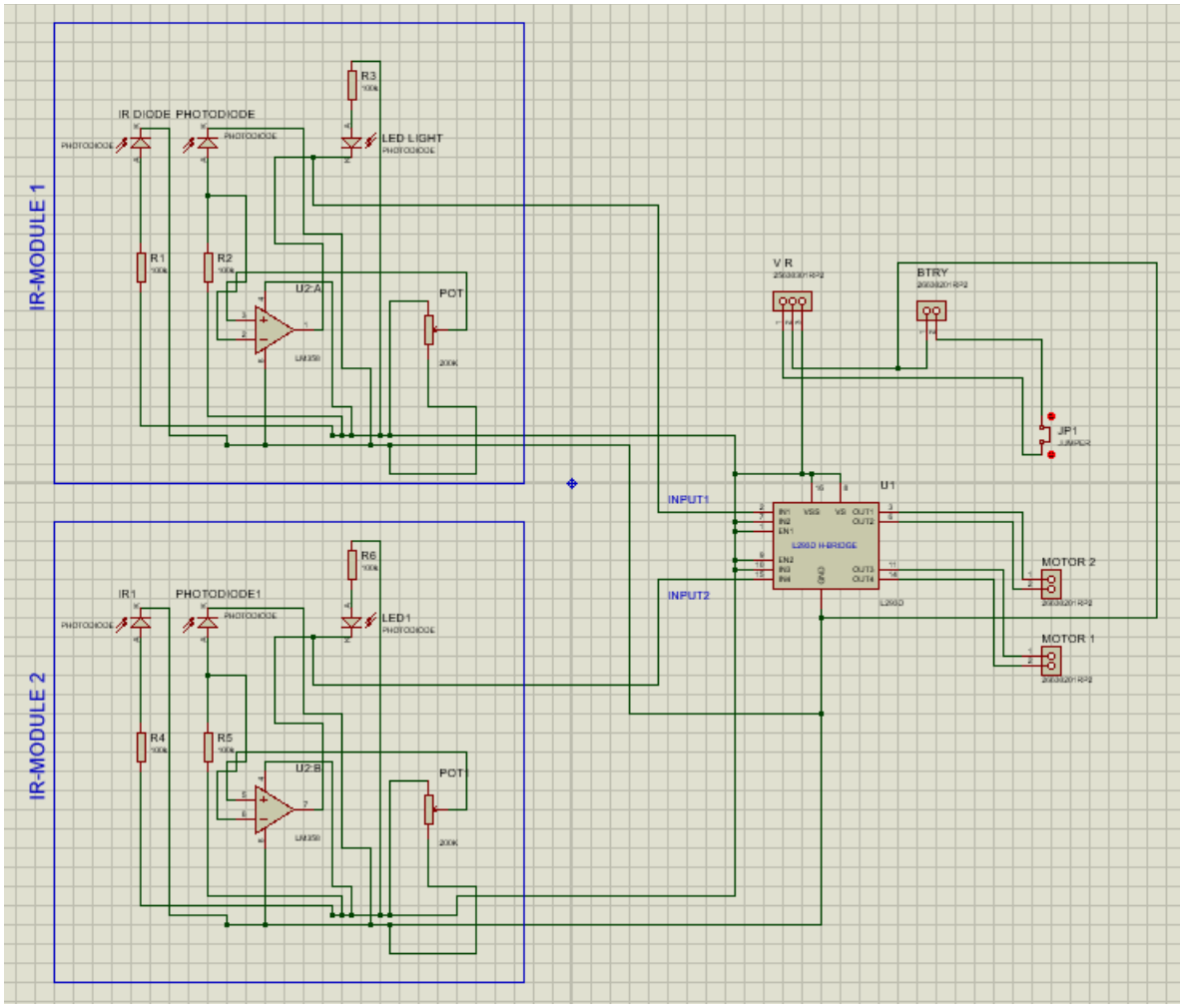


Figure 3.2 Schematic Diagram of the Circuit

3.3 Working Mechanism

The circuit that powers our IR sensors and our L293D motor driver is set up using a rechargeable battery and a voltage regulator. For our circuit, we use two IR sensors. When an IR sensor detects a white path, it gives the circuit a high input, but when it detects a black path, it gives the circuit a LOW input. The LM358 IC, an 8-pin IC with two OP-amps, a potentiometer, an IR LED and LED, a 100K Ohm resistance, a 100 Ohm

resistance, and a battery connector, make up the IR sensors. The IR sensors' width must be only a hair wider than the black line's width.

When the IR sensors are coupled to a voltage regulator and a battery, the sensors will get a constant voltage. The IR LED and LED, which serve as an IR transmitter and receiver, are used by IR sensors to detect the white path. When both IR sensors detect a white path, a signal is delivered to pins 2, 3, and 5 and pins 1 and 7 of the LM358 IC, which transforms the input and provides a HIGH output to the L293D motor driver. When the LM358 IC's two output pins both deliver the HIGH output to the motor driver, the two motors are operated simultaneously from both output signals, and the robot advances. And the robot stops when the IR sensors detect a black passage by sending the motor driver a LOW signal.

When one of the IR sensors detects a black path, the motor driver receives LOW output from one of the IC's output pins for turning movement, while receiving HIGH output from another pin. As a result, one of the motors continues to run while the other motor is inactive, causing the robot to rotate around its axis and achieving rotational movement.

A switch is also included so that the robot can be stopped at any time. In addition, two LM741 IC, an 8 pin 1 OP-amp IC, can be utilized if LM358 ICs are not readily available.

3.4 Hardware Required:

3.4.1 DC Motor

It is used on the rare two wheels of the line following robot. These motors can operate



Figure 1:DC Motor

within the range of 5-9V.

3.4.2 L293D H-Bridge Motor Driver

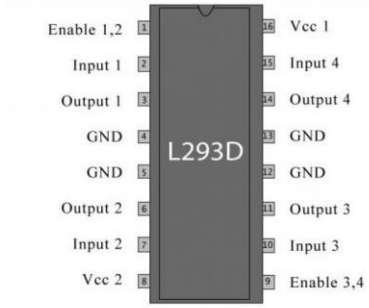


Figure 2:L293D Motor Driver

This component is used to control the two DC motors of the rare wheels simultaneously enabling the robot to make turns in the required directions.

3.4.3 LM358 IC



Figure 3:LM358 IC

It is used as a comparator which compares two input signals coming from IR sensors and gives one output signal to the motor driver.

3.4.4 Rechargeable Battery 9V



Figure 4:Rechargeable battery

A 9V rechargeable battery is used for providing power supply to the line following robot.

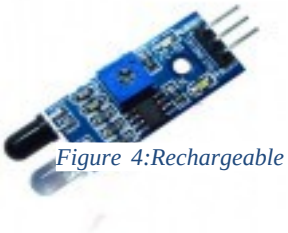


Figure 5:IR Sensor

3.4.5 Sensor Module(IR LED, Photodiode, Potentiometer)

Two IR sensors are used to sense the path and provide signals to the motor driver.

3.4.6 Switch



Figure 6:Switch

A switch is used in the robot to stop and begin movement of the robot.

3.4.7 Front Wheel



Figure 7:Front Wheel

It helps to maintain stability and balance of the robot specially at turns or curves.

3.4.8 Back Wheel



Figure 8: Back Wheel

The two back wheels are responsible for moving the robot forward and making turns based on instructions from the motor drivers.

3.4.9 Chassis

Chassis provides structural support, protection and helps to distribute the weight of robot evenly throughout the robot.



Figure 9:Chassis

3.4.10 Voltage Regulator

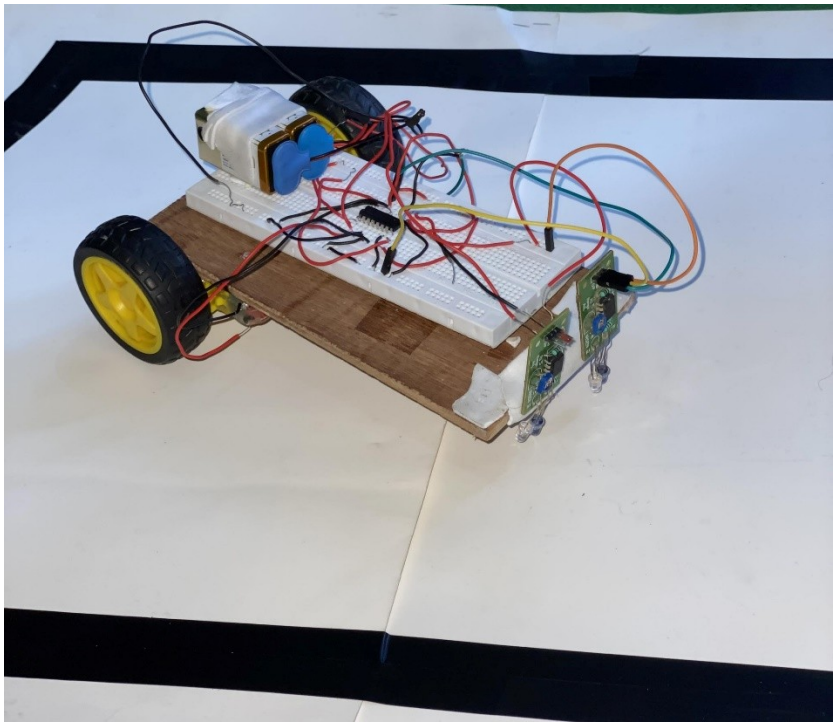


Figure 10:Voltage Regulator

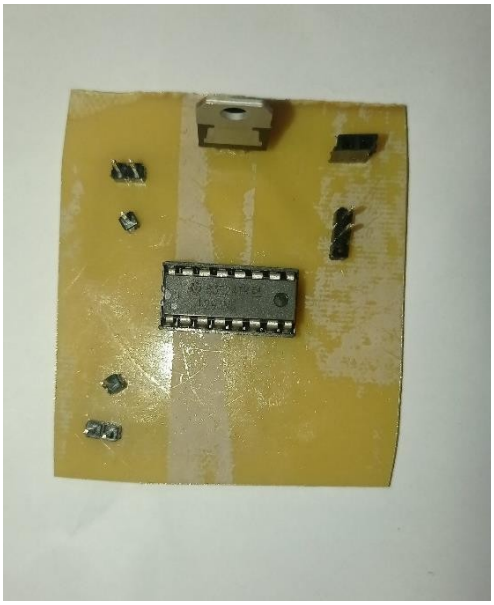
It stabilizes the voltage enabling the IR sensors to work efficiently which helps the robot to follow the course accurately.

3.5 Design

1. Robot Body:



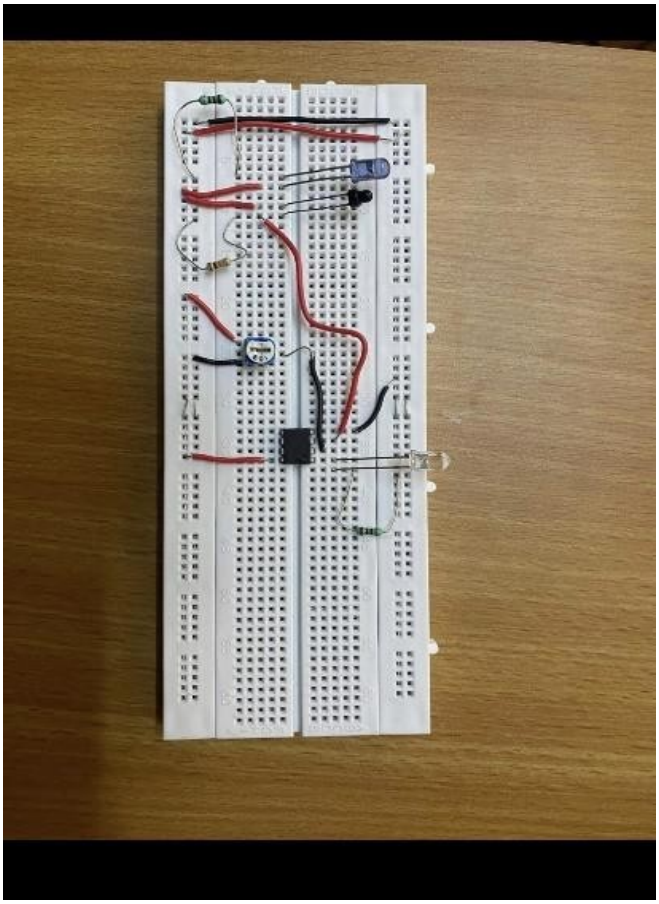
2. Arrangement Sensors:



3. Design of path:



4. Circuit Simulation



5. PCB Design

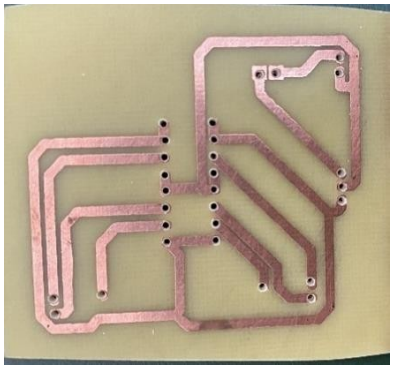


Fig-PCB design of motordriver

CHAPTER 4: RESULT AND ANALYSIS

4.1 Practical Data Analysis

Throughout the project, practical data analysis were done. A single 9V battery is unable to run the motor and IR sensor simultaneously due to less power. So, double 9V battery in parallel circuit is used.

Data for voltage regulator for different voltage supplied

Voltage Supplied	Voltage provided	Current provided
12V	5V	1.5A
9V	5V	1.5A

Data obtained by IR Sensors when 5V is supplied

Voltage Across	Voltage Obtained LED is on	Voltage Obtained LED is off
IR LED	1.21V	1.21V
Photodiode	0.22V	2.25V
Output to Pin 1 IC	3.57V	0V

4.2 Cost Estimation

S.N.	Components Name	Quantity	Cost	Total Cost
1.	DC Motor	2	150	300
2.	L239D Motor Driver	1	275	275
3.	LM358 IC	1	50	50
4.	Jumper Wire (Packet)	1	140	140
5.	Rechargeable Battery	1	600	600
6.	Voltage Regulator	1	100	100
7.	Sensor Module (IR LEDs, potentiometer, resistors)	2	325	650
8.	Switch	1	60	60
9.	Back wheels	2	100	200
10.	Front wheels	1	100	100
11.	Chassis	1	100	100
	TOTAL COST			2575

CHAPTER 5: CONCLUSIONS

5.1 Conclusion

In order to stay on track with our marked black path, LFR is an autonomous system that detects the way by sending and receiving infrared waves from the IR LED. The project, which was based on the feedback of two ICs—the LM358 IC and the L293D IC. While it was difficult to finish this project in three months, it helped us learn more in a short period of time and highlighted the importance of teamwork and communication to project success. The project's completion is an accomplishment for all of us and will motivate us to work harder in the days ahead.

5.2 Future Works

The following future works that can be carried out to modify this project:

i. Use of microcontrollers in order for smooth operation of the robot (Example: To alter the speed).
ii. Use of ultrasonic sensors in order to detect obstacle on the path.
iii. Use of multiple IR sensors to make the robot go backwards.

GANTT CHART

The following shows the timeline of the project

ACTIVITY	AUG	SEP	OCT	NOV	DEC
PROPOSAL DEFENSE					
LITERATURE REVIEW					
CIRCUIT DESIGN					
PCB MAKING					
ASSEMBLY					
TESTING AND MODIFICATION					
REPORT MAKING					
FINAL PRESENTATION					

WORK COMPLETED	

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APPENDIX

1. Datasheet of IC

ELECTRICAL CHARACTERISTICS (V_{CC} = 5.0 V, V_{EE} = Gnd, T_A = 25°C, unless otherwise noted.)

Characteristic	Symbol	LM268			LM358			LM2904			LM2904V			Unit
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage V_{CC} = 5.0 V to 30 V (26 V for LM2904, V), V_{IC} = 0 V to V_{CC} −1.7 V, V_O = 1.4 V, R_S = 0 Ω	V_{IO}													mV
T_A = 25°C		–	2.0	5.0	–	2.0	7.0	–	2.0	7.0	–	–	–	
T_A = T_{high} (Note 1)		–	–	7.0	–	–	9.0	–	–	10	–	–	13	
T_A = T_{low} (Note 1)		–	–	2.0	–	–	9.0	–	–	10	–	–	10	
Average Temperature Coefficient of Input Offset Voltage T_A = T_{high} to T_{low} (Note 1)	$\Delta V_{IO}/\Delta T$	–	7.0	–	–	7.0	–	–	7.0	–	–	7.0	–	μV/°C
Input Offset Current	I_{IO}	–	3.0	30	–	5.0	50	–	5.0	50	–	5.0	50	nA
T_A = T_{high} to T_{low} (Note 1)		–	–	100	–	–	150	–	45	200	–	45	200	
Input Bias Current	I_{IB}	–	−45	−150	–	−45	−250	–	−45	−250	–	−45	−250	
T_A = T_{high} to T_{low} (Note 1)		–	−50	−300	–	−50	−500	–	−50	−500	–	−50	−500	
Average Temperature Coefficient of Input Offset Current T_A = T_{high} to T_{low} (Note 1)	$\Delta I_{IO}/\Delta T$	–	10	–	–	10	–	–	10	–	–	10	–	pA/°C
Input Common Mode Voltage Range (Note 2), V_{CC} = 30 V (26 V for LM2904, V)	V_{ICR}	0	–	28.3	0	–	28.3	0	–	24.3	0	–	24.3	V
V_{CC} = 30 V (26 V for LM2904, V), T_A = T_{high} to T_{low}		0	–	28	0	–	28	0	–	24	0	–	24	
Differential Input Voltage Range	V_{IDR}	–	–	V_{CC}	–	–	V_{CC}	–	–	V_{CC}	–	–	V_{CC}	V
Large Signal Open Loop Voltage Gain R_L = 2.0 kΩ, V_{CC} = 15 V, For Large V_O Swing,	A_{VOL}	50	100	–	25	100	–	25	100	–	25	100	–	V/mV
T_A = T_{high} to T_{low} (Note 1)		25	–	–	15	–	–	15	–	–	15	–	–	
Channel Separation 1.0 kHz ≤ f ≤ 20 kHz, Input Referenced	CS	–	−120	–	–	−120	–	–	−120	–	–	−120	–	dB
Common Mode Rejection R_S ≤ 10 kΩ	CMR	70	85	–	65	70	–	50	70	–	50	70	–	dB
Power Supply Rejection	PSR	65	100	–	65	100	–	50	100	–	50	100	–	dB
Output Voltage−High Limit (T_A = T_{high} to T_{low}) (Note 1)	V_{OH}													V
V_{CC} = 5.0 V, R_L = 2.0 kΩ, T_A = 25°C		3.3	3.5	–	3.3	3.5	–	3.3	3.5	–	3.3	3.5	–	
V_{CC} = 30 V (26 V for LM2904, V), R_L = 2.0 kΩ		26	–	–	26	–	–	22	–	–	22	–	–	
V_{CC} = 30 V (26 V for LM2904, V), R_L = 10 kΩ		27	28	–	27	28	–	23	24	–	23	24	–	
Output Voltage−Low Limit V_{CC} = 5.0 V, R_L = 10 kΩ, T_A = T_{high} to T_{low} (Note 1)	V_{OL}	–	5.0	20	–	5.0	20	–	5.0	20	–	5.0	20	mV
Output Source Current V_{ID} = +1.0 V, V_{CC} = 15 V	I_{O+}	20	40	–	20	40	–	20	40	–	20	40	–	mA
Output Sink Current	I_{O-}													
V_{ID} = −1.0 V, V_{CC} = 15 V		10	20	–	10	20	–	10	20	–	10	20	–	mA
V_{ID} = −1.0 V, V_O = 200 mV		12	50	–	12	50	–	–	–	–	–	–	–	μA
Output Short Circuit to Ground (Note 3)	I_{SC}	–	40	60	–	40	60	–	40	60	–	40	60	mA
Power Supply Current (T_A = T_{high} to T_{low}) (Note 1)	I_{CC}													mA
V_{CC} = 30 V (26 V for LM2904, V), V_O = 0 V, R_L = ∞		–	1.5	3.0	–	1.5	3.0	–	1.5	3.0	–	1.5	3.0	
V_{CC} = 5 V, V_O = 0 V, R_L = ∞		–	0.7	1.2	–	0.7	1.2	–	0.7	1.2	–	0.7	1.2	

NOTES: 1. T_{low} = −40°C for LM2904
= −40°C for LM2904V
= −25°C for LM258
= 0°C for LM358

T_{high} = +105°C for LM2904
= +125°C for LM2904V
= +85°C for LM258
= +70°C for LM358

2. The input common mode voltage or either input signal voltage should not be allowed to go negative by more than 0.3 V. The upper end of the common mode voltage range is V_{CC} −1.7 V.
3. Short circuits from the output to V_{CC} can cause excessive heating and eventual destruction. Destructive dissipation can result from simultaneous shorts on all amplifiers.

Datasheet of LM358 IC

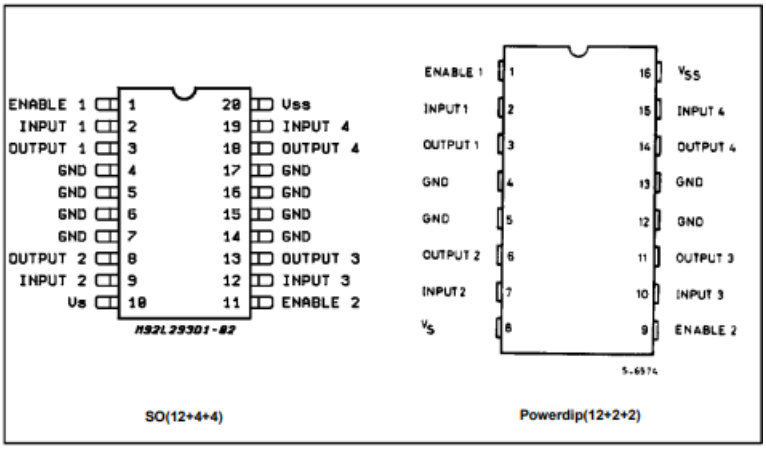
2. Connections for L293D

L293D - L293DD

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V_S	Supply Voltage	36	V
V_{SS}	Logic Supply Voltage	36	V
V_i	Input Voltage	7	V
V_{en}	Enable Voltage	7	V
I_o	Peak Output Current (100 μs non repetitive)	1.2	A
P_{tot}	Total Power Dissipation at T_{pins} = 90 °C	4	W
T_{stg}, T_j	Storage and Junction Temperature	– 40 to 150	°C

PIN CONNECTIONS (Top view)



SO(12+4+4) Powerdip(12+2+2)

THERMAL DATA

Symbol	Decription		DIP	SO	Unit
$R_{th\ j\text{-}pins}$	Thermal Resistance Junction-pins	max.	–	14	°C/W
$R_{th\ j\text{-}amb}$	Thermal Resistance junction-ambient	max.	80	50 (*)	°C/W
$R_{th\ j\text{-}case}$	Thermal Resistance Junction-case	max.	14	–	

(*) With 6sq. cm on board heatsink.